

INDIAN INSTITUTE OF TECHNOLOGY MADRAS

CV5570: Computer Vision for Construction Engineering

Assignment 2 - Report

Comparative Evaluation of Faster R-CNN, RetinaNet, and RT-DETR-L for Construction Equipment Detection

Submitted by:
N P J Vedhaanth
Roll Number: ID26S422

GitHub Repository: <https://github.com/wick1-22/Assignment-2-CV5570>

ACID dataset | Classes: Dozer, Dump Truck, Excavator

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1. Executive Summary

This work compares a two-stage detector (Faster R-CNN), a one-stage detector (RetinaNet), and a transformer detector (RT-DETR-L) on the ACID construction-image subset. The experiment uses 3,929 images and 7,614 annotated objects from three classes. The highest validation mAP@0.5:0.95 was RT-DETR-L (0.750); the highest measured speed was RetinaNet (16.8 FPS). The choice between accuracy and throughput is discussed with the per-class scores and example detections.

2. Objective and Scope

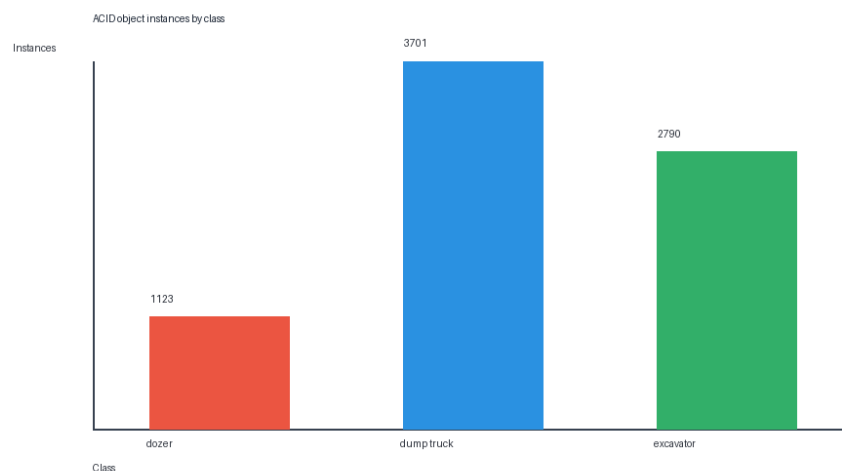
The objective was to train and compare one detector from each of the two-stage, one-stage, and transformer families. The comparison covers validation AP at IoU 0.5 and 0.50:0.95, class-wise AP, inference speed, parameter count, training time, convergence, and qualitative successes and failures. The test split was held out.

3. Dataset Description and Preprocessing

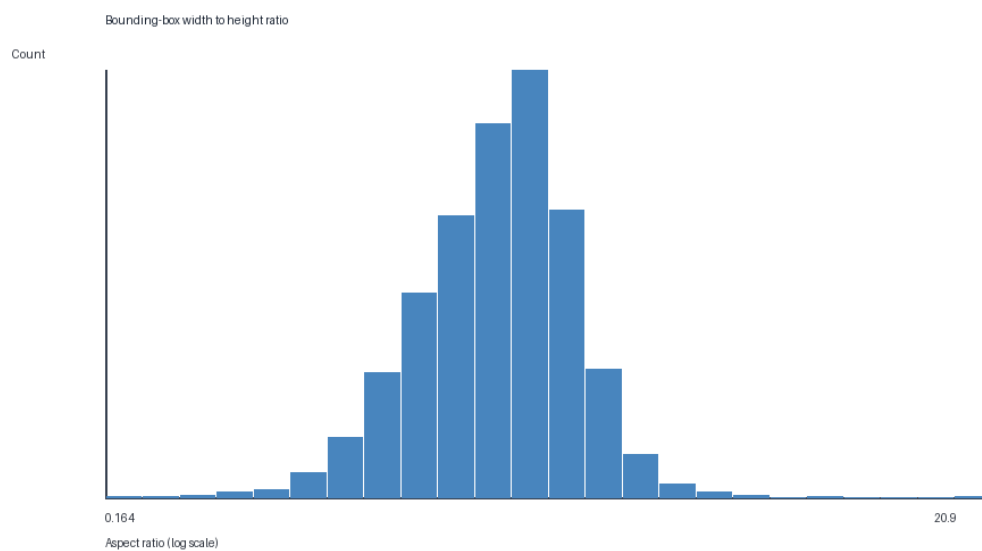
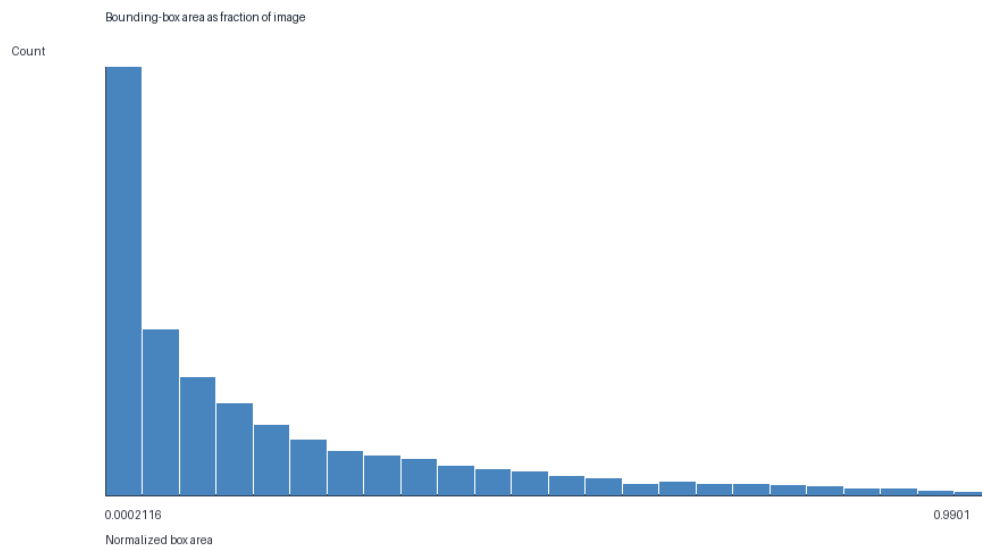
The selected ACID subset contains 3,929 images and 7,614 object instances. The annotations are COCO JSON, with bounding boxes stored in pixel coordinates as [x, y, width, height]. Image references were checked before processing. The images were shuffled using seed 42 and split into 70% training, 20% validation, and 10% test.

Split	Images	Instances	Dozer	Dump Truck	Excavator
Train	2750	5351	776	2614	1961
Val	785	1499	243	715	541
Test	394	764	104	372	288

The full class counts are Dozer 1,123, Dump Truck 3,701, and Excavator 2,790. Median normalized box area is 0.088; median width-to-height ratio is 1.45.



Dataset distributions and examples



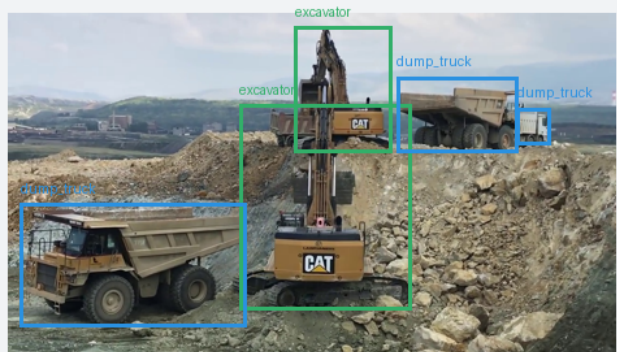
Dataset examples

Six dataset images with their ground-truth boxes are shown below. These are dataset examples, not model predictions.

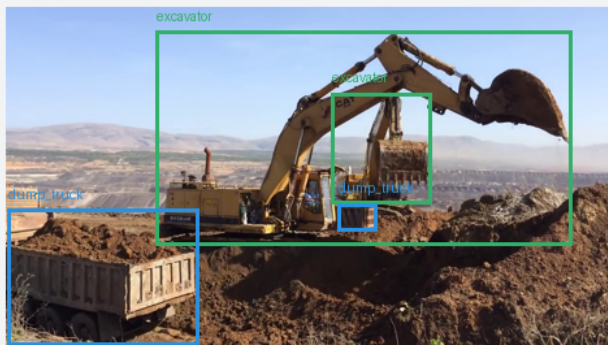
Ground truth: 05181.jpg



Ground truth: 03222.jpg



Ground truth: 06435.jpg



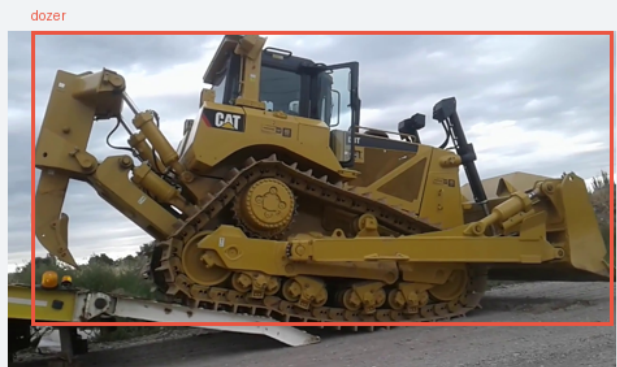
Ground truth: 03025.jpg



Ground truth: 00538.jpg



Ground truth: 00615.jpg



4. Detector Methods and Training Setup

Faster R-CNN uses a region proposal network to identify candidate object regions, followed by a classifier and box-refinement stage. It was selected as a two-stage baseline for equipment at different sizes and against cluttered site backgrounds (Ren et al., 2015). RetinaNet predicts class scores and box offsets densely in one pass; focal loss reduces the weight of easy background examples, a common source of imbalance in detection (Lin et al., 2017). RT-DETR-L uses a transformer encoder/decoder and set-based object predictions. It was included to compare the transformer family on these scenes and measure its speed/accuracy trade-off (Zhao et al., 2024). The predictor used for this run does not return attention maps, so its errors are reviewed using validation predictions and ground truth.

Faster R-CNN and RetinaNet use TorchVision COCO-pretrained ResNet-50-FPN v2 weights. RT-DETR-L uses the pretrained `rtdetr-l` checkpoint. The three runs use the same saved split, input size, epoch budget, and disabled augmentations. Their training libraries and optimizer settings are not identical.

Model	Pretrained checkpoint	Optimizer	Learning rate	Batch	Epochs	Device
Faster R-CNN	Faster R-CNN ResNet50-FPN v2 COCO weights	SGD	0.0050	2	10	cuda
RetinaNet	RetinaNet ResNet50-FPN v2 COCO weights	SGD	0.0005	2	10	cuda
RT-DETR-L	rtdetr-l.pt	AdamW	0.0014	2	10	cuda:0

Faster R-CNN used SGD with learning rate 0.005, momentum 0.9, and weight decay 0.0005. RetinaNet used learning rate 0.0005 after the initial 0.005 attempt produced non-finite losses; momentum and weight decay were unchanged. RT-DETR used the optimizer settings recorded by Ultralytics. Training was run on the device recorded in each model's metrics file.

Runtime versions: Python Google Colab Python 3.13 (reported by runtime), PyTorch version not saved in run artifacts, TorchVision version not saved in run artifacts, Ultralytics version not saved in run artifacts. GPU: NVIDIA Tesla T4.

5. Validation Results

AP is calculated with 101-point interpolated precision-recall. FPS is measured on the validation images using the run's inference path and device.

Model	Family	mAP@0.5	mAP@0.5:0.95	FPS	Parameters (M)	Training time (min)
Faster R-CNN	two-stage	0.929	0.679	9.6	43.27	138.7
RetinaNet	one-stage	0.896	0.625	16.8	36.37	79.3
RT-DETR-L	transformer	0.939	0.750	13.0	32.81	72.3

Model	Class	AP@0.5	AP@0.5:0.95
Faster R-CNN	Dozer	0.937	0.691
Faster R-CNN	Dump Truck	0.896	0.617
Faster R-CNN	Excavator	0.953	0.729
RetinaNet	Dozer	0.939	0.674
RetinaNet	Dump Truck	0.817	0.549
RetinaNet	Excavator	0.933	0.654
RT-DETR-L	Dozer	0.948	0.754
RT-DETR-L	Dump Truck	0.910	0.677
RT-DETR-L	Excavator	0.958	0.818

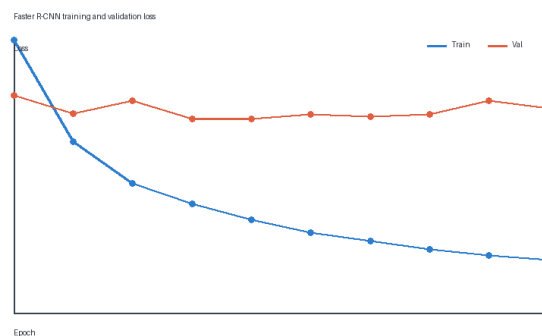


6. Training and Convergence

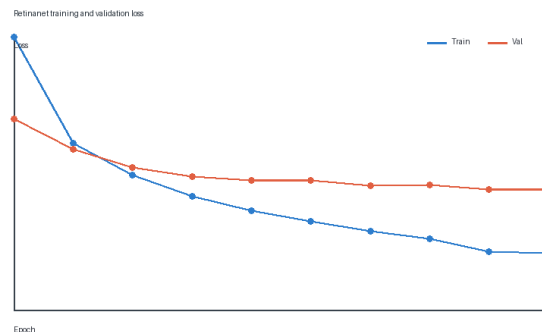
The training and validation curves show how the recorded objective changed across epochs. Loss values have model-specific definitions and should be compared as trends within each model, not as directly equivalent scales.

Faster R-CNN: its validation objective changed by 5.9% from epoch 1 to the last epoch and was flattening or changing direction near the end. RetinaNet: its validation objective changed by 37.2% from epoch 1 to the last epoch and was still decreasing at the end. RT-DETR-L: its validation objective changed by 27.4% from epoch 1 to the last epoch and was still decreasing at the end. The minimum recorded validation loss occurred at epoch Faster R-CNN 5, RetinaNet 10, RT-DETR-L 10. For 10 epochs, recorded training time was Faster R-CNN 138.7 min, RetinaNet 79.3 min, RT-DETR-L 72.3 min. RT-DETR-L reached its lowest recorded validation loss at epoch 10, compared with epoch 5 for Faster R-CNN and epoch 10 for RetinaNet. Its 72.3-minute run was shorter in wall time than Faster R-CNN (138.7 min) and RetinaNet (79.3 min); loss definitions differ, so this compares convergence timing rather than loss magnitude. Loss definitions differ across architectures, so compare their within-run trends and time-to-best rather than the absolute loss values.

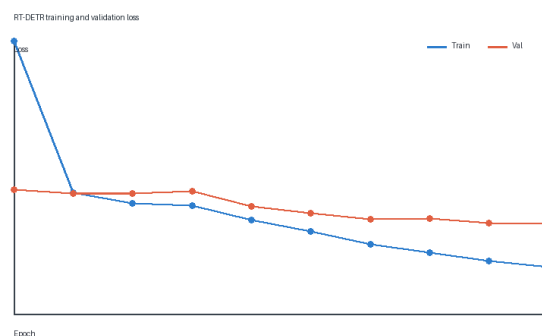
Faster R-CNN



RetinaNet



RT-DETR-L



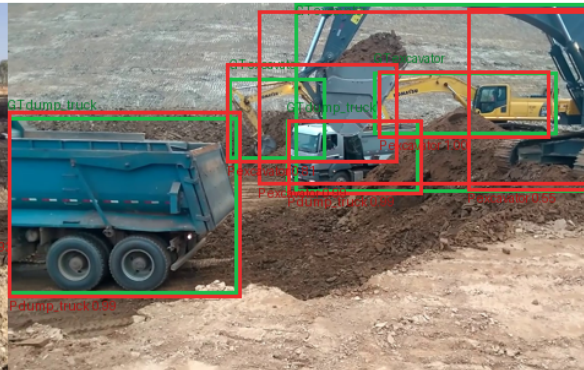
Each board uses validation images. Green boxes are ground truth and red boxes are predictions. Success examples contain a correct match with at least 50% recall and 50% precision at IoU 0.5; the displayed prediction threshold is 0.25. Failure examples use the same rule and are not swapped for easier images.

Faster R-CNN validation examples

Success: 00002.jpg | TP 2 FP 2 FN 0



Success: 00018.jpg | TP 4 FP 2 FN 1



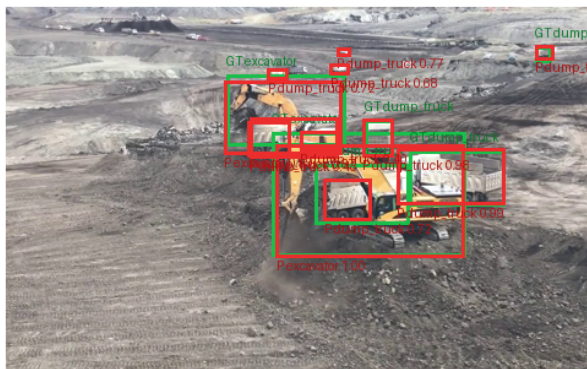
Success: 00028.jpg | TP 1 FP 1 FN 0



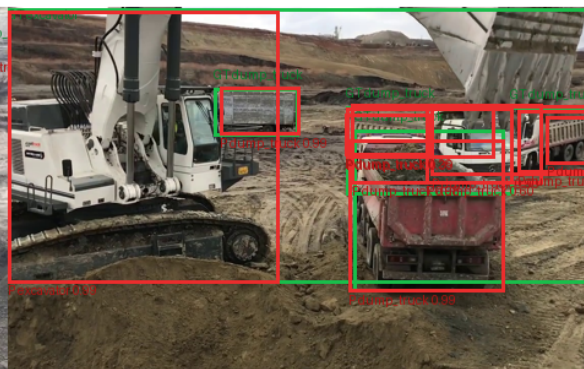
Failure: 04460.jpg | TP6 FP9 FN 2



Failure: 05513.jpg | TP6 FP7 FN 1



Failure: 09385.jpg | TP5 FP6 FN 1



Type	Image	TP	FP	FN
Success	00002.jpg	2	2	0
Success	00018.jpg	4	2	1
Success	00028.jpg	1	1	0
Failure	04460.jpg	6	9	2
Failure	05513.jpg	6	7	1
Failure	09385.jpg	5	6	1

RetinaNet validation examples

Success: 00002.jpg | TP 2 FP 1 FN 0



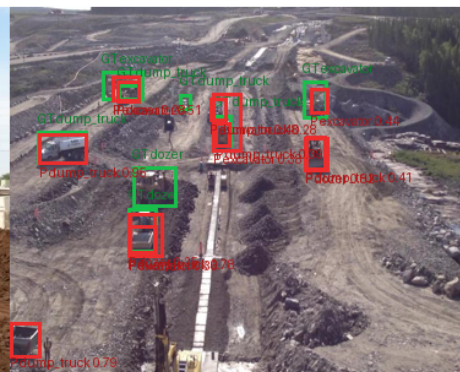
Success: 00018.jpg | TP 4 FP 2 FN 1



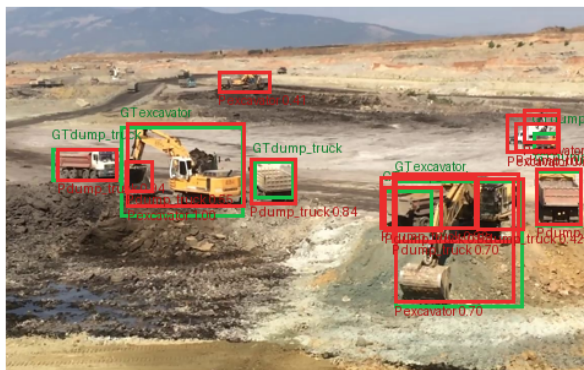
Success: 00028.jpg | TP 1 FP 0 FN 0



Failure: 07480.jpg | TP 4 FP 10 FN 4



Failure: 08970.jpg | TP6 FP9 FN 2



Failure: 05513.jpg | TP5 FP9 FN 2



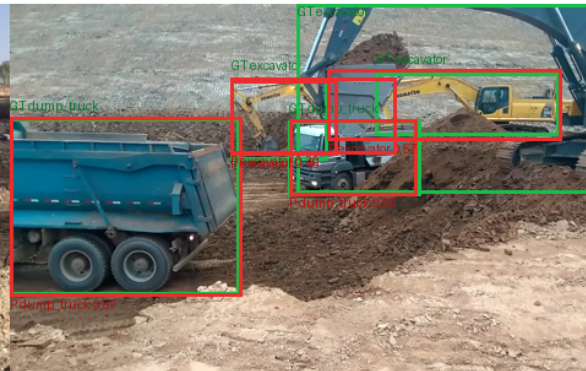
Type	Image	TP	FP	FN
Success	00002.jpg	2	1	0
Success	00018.jpg	4	2	1
Success	00028.jpg	1	0	0
Failure	07480.jpg	4	10	4
Failure	08970.jpg	6	9	2
Failure	05513.jpg	5	9	2

RT-DETR-L validation examples

Success: 00002.jpg | TP 2 FP 1 FN 0



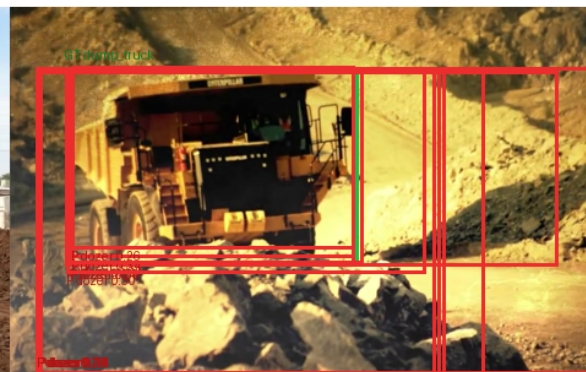
Success: 00018.jpg | TP 4 FP 1 FN 1



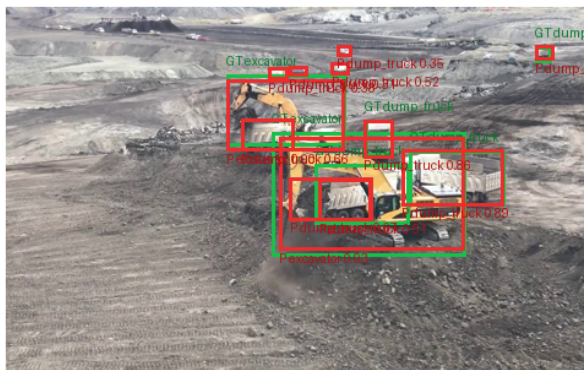
Success: 00028.jpg | TP 1 FP 0 FN 0



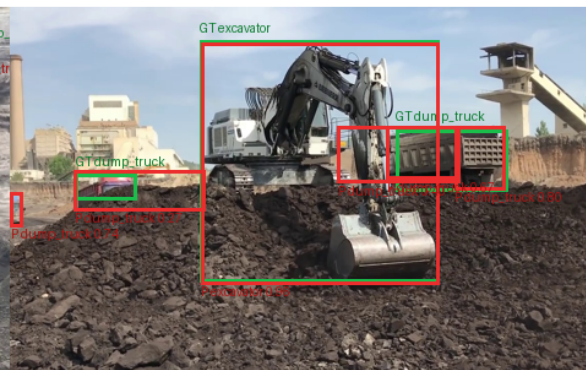
Failure: 06108.jpg | TP 0 FP 11 FN 1



Failure: 05513.jpg | TP 6 FP 7 FN 1



Failure: 02159.jpg | TP 1 FP 5 FN 2



Type	Image	TP	FP	FN
Success	00002.jpg	2	1	0
Success	00018.jpg	4	1	1
Success	00028.jpg	1	0	0
Failure	06108.jpg	0	11	1
Failure	05513.jpg	6	7	1
Failure	02159.jpg	1	5	2

8. Comparison and Deployment Discussion

The fastest model in this run was RetinaNet at 16.8 FPS. The highest mAP@0.5:0.95 was from RT-DETR-L (0.750). The fastest model is the initial candidate for real-time site monitoring, while the most accurate is the initial candidate for offline batch analysis. This depends on whether its class-wise scores and failure cases meet the use-case requirements.

Observed in the displayed validation examples: Faster R-CNN detects the excavator in 00002.jpg (TP 2, FP 2, FN 0) and the dozer in 00028.jpg (TP 1, FP 1, FN 0), while the crowded or overlapping scenes 04460.jpg, 05513.jpg, and 09385.jpg contain extra predictions and missed objects. RetinaNet detects the isolated dozer in 00028.jpg, but the crowded scenes 07480.jpg and 08970.jpg contain several false boxes and missed objects. RT-DETR produces no correct match for the dump truck in 06108.jpg (11 false positives and one missed object); the crowded scenes 05513.jpg and 02159.jpg also show overlapping extra boxes and missed instances.

Possible explanations to investigate: small objects at 640-pixel input resolution may be harder to localize; CNN anchor sizes may not fit the observed box shapes; the class distribution is imbalanced; and overlap or NMS may contribute to duplicate or suppressed detections. These are hypotheses, not causes established by this experiment.

9. Limitations

The experiment uses a single random image split rather than grouping images by scene, so similar views may appear in more than one subset. The epoch budget is short, and the model families use different training implementations and optimization settings. FPS depends on the GPU, software versions, and inference path. The reported metrics are validation results; the held-out test split was not used for the comparison.

10. Conclusion

On this validation split, RT-DETR-L achieved the highest mAP@0.5:0.95 (0.750), while RetinaNet had the highest measured speed (16.8 FPS). These measurements give an accuracy-oriented and a speed-oriented option. The selected class-wise AP and failure examples should guide the final choice for a specific monitoring or batch-analysis system.

11. References

- [1] Ren, S. et al. 'Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks.' NeurIPS, 2015. https://papers.nips.cc/paper_files/paper/2015/hash/14bfa6bb14875e45bba028a21ed38046-Abstract.html
- [2] Lin, T.-Y. et al. 'Focal Loss for Dense Object Detection.' ICCV, 2017. https://openaccess.thecvf.com/content_ICCV_2017/papers/Lin_Focal_Loss_for_Dense_Object_Detection_ICCV_2017_paper.pdf
- [3] Zhao, Y. et al. 'DETRs Beat YOLOs on Real-time Object Detection.' CVPR, 2024. https://openaccess.thecvf.com/content/CVPR2024/html/Zhao_DETRs_Beat_YOLOs_on_Real-time_Object_Detection_CVPR_2024_paper.html
- [4] ACID dataset: link provided in the CV5570 Assignment 2 brief. https://drive.google.com/uc?id=1Qg_X5FygUMBRTcVFPb0s1fQP-20f8n8O
- [5] Ultralytics. 'RT-DETR.' <https://docs.ultralytics.com/models/rtdetr>